

# Faithful RAG: Evaluating Citation Accuracy and Retrieval Robustness in Domain-Specific Scientific Literature

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Team Faithful RAG

## 1 Introduction

Retrieval-Augmented Generation (RAG) has become the standard approach for grounding LLM outputs in external knowledge (Lewis et al., 2020), yet citation presence does not guarantee citation accuracy: models frequently cite the wrong passage, attribute claims to non-existent sources, or contradict their own retrieved context (Min et al., 2023). This problem is especially acute in scientific and policy domains, where a fabricated citation cannot be verified without reading the original paper.

We address this in the domain of **intellectual property and innovation policy**. Building on an existing RAG pipeline that ingests academic papers via VLM-based PDF parsing, we design a systematic evaluation of citation faithfulness and retrieval robustness. Our contributions are: (1) a domain-specific gold benchmark of 50 expert-annotated question-answer pairs with ground-truth passages; (2) an NLI-based citation verification pipeline following the ALCE framework (Gao et al., 2023); (3) an implementation of Corrective RAG (Yan et al., 2024) with iterative query refinement inspired by FLARE (Jiang et al., 2023); and (4) a multi-axis retrieval ablation comparing bi-encoders and late-interaction models.

## 2 Methodology

**RAG Pipeline.** We ingest academic PDFs using a vision-language model (Dolphin 1.5) for layout-aware markdown extraction, apply hybrid semantic chunking (coarse  $\sim 2000$  and fine  $\sim 300$  chars), embed with FAISS, optionally rerank with a neural reranker, and generate cited answers via multiple LLMs.

**Gold Dataset.** Each team member annotates 12–13 question-answer pairs from the corpus with ground-truth passages at chunk level, spanning policy impact, methodology, cross-country comparison, factual extraction, and multi-hop reasoning.

**Retrieval Ablation.** We compare four retrieval approaches—three bi-encoders (OpenAI text-embedding-3-small, BGE-M3, E5-large) and ColBERTv2 (Santhanam et al., 2022) for late interaction—across chunk granularities with and without reranking, measuring precision@k, recall@k, MRR, and hit rate.

**Citation Verification.** We decompose generated answers into atomic claims via prompted extraction, then classify each claim–citation pair using DeBERTa-v3-large-mnli for NLI, labeling claims as *supported*, *not supported*, *fabricated source*, or *no citation*.

**Corrective RAG.** A retrieval quality evaluator scores document relevance after initial retrieval. When confidence falls below a threshold, the system reformulates the query and re-retrieves. We ablate the confidence threshold.

**End-to-End Evaluation.** We evaluate using RAGAS metrics (faithfulness, answer relevancy, context precision, context recall) and compare chunked RAG against a long-context baseline (128K window).

## 3 Timeline and Team Roles

**Weeks 1–2** (Apr 21–May 3): Repository setup, gold dataset annotation, proposal submission.

**Weeks 3–4** (May 5–18): Parallel implementation of all four evaluation components.

**Week 5** (May 19–24): Preliminary experiments; submit progress report and notebooks.

**Weeks 6–7** (May 26–Jun 7): Full experiments, ablations, final report.

**Elie Bruno:** Retrieval ablation (embeddings, chunking, reranking).

**Andrea Trugenberger:** Citation verification (NLI pipeline, cross-LLM comparison).

**Member 3:** Corrective RAG implementation and threshold ablation.

**Member 4:** RAGAS evaluation and long-context baseline.

All members contribute to the gold evaluation dataset.

## References

- Tianyu Gao, Howard Yen, Jiatong Yu, and Danqi Chen. 2023. [Enabling large language models to generate text with citations](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.
- Zhengbao Jiang, Frank F. Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. 2023. [Active retrieval augmented generation](#).
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. [Retrieval-augmented generation for knowledge-intensive NLP tasks](#).
- Sewon Min, Kalpesh Krishna, Xinxì Lyu, Mike Lewis, Wen tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. 2023. [FActScore: Fine-grained atomic evaluation of factual precision in long form text generation](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*.
- Keshav Santhanam, Omar Khattab, Jon Saad-Falcon, Christopher Potts, and Matei Zaharia. 2022. [ColBERTv2: Effective and efficient retrieval via lightweight late interaction](#). In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics*.
- Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. 2024. [Corrective retrieval augmented generation](#).